When do kernel tests raise false alarms on long-memory data?

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Abstract

Kernel tests were built for independent observations, but many ordered data have long memory. We measure when kernel tests raise false alarms on such data, and we give one decision rule for testing independence. Remove the seasonal mean and scale from each series. Then run the Hilbert–Schmidt independence criterion (HSIC) with circular shifts, which realign one series against the other to build the reference distribution. In our main simulation arms at $n = 2000$, this rule rejected 3.8%–7.0% of true nulls at a nominal 5%. At matched size it kept the power of the best other HSIC rule. The rule has limits. No long-memory theory covers it, a slow scale drift raises it to 10.1%, and under strong memory its rate creeps up with n. Two findings support the rule. First, the critical-value rule sets the false-alarm rate. Split-sample maximum mean discrepancy (MMD) with wild multipliers rejects 27.1% of stationary series at Hurst index $H = 0.8$. Second, a seasonal scale shared by independent series drives HSIC to 73.3% without memory, while Pearson correlation stays at 4.7%. We prove that a shared seasonal mean with random phase drives the size of common rules to one. On Nasdaq trade durations the rejection fractions remain descriptive.

Keywords: long-range dependence; kernel tests; Hermite rank; MMD; HSIC; distance covariance; seasonality; market microstructure.

1 Introduction

Maximum mean discrepancy (MMD) [Gretton et al., 2012] is often used to flag a change in distribution. The Hilbert–Schmidt independence criterion (HSIC) [Gretton et al., 2007] flags dependence between paired variables. Distance covariance [Székely et al., 2007] asks the same question as HSIC through pairwise distances. All three tests were introduced for independent and identically distributed observations. A nominal 5% test should reject a true null about five times in 100 repeated samples. On persistent data that promise can fail if the critical value ignores the dependence. This paper asks when the promise fails under long memory, and what an analyst should do instead. For independence testing our answer is a three-step rule. Remove the seasonal mean and the seasonal scale from each series. Then test with HSIC and circular shifts. A circular shift realigns one full series against the other by a wrapped offset, and the shifted statistics form the reference distribution. Finally, expect residual risk from slow drift that the adjustment cannot remove. Box 1 at the end of this section states the rule, the evidence behind it, and its limits.

The paper makes three contributions. First, it measures false-alarm rates under known null models and shows that the critical-value rule matters as much as the statistic. At $n = 2000$, split-sample MMD with uncentred wild multipliers rejects 27.1% of 1,000 stationary fractional Gaussian noise (fGn) series at $H = 0.8$. HSIC with subsampling rejects 9.2% of 2,000 independent fGn pairs at $H_X = H_Y = 0.7$. A Hermite-rank calculation explains why the one-series split and the two-series product fail at different memory levels. An exact calculation shows that memory widens the null spread of the half-mean difference, so a fixed mean shift becomes hard to detect for any split test. Second, a factorial experiment separates memory from heavy tails, nonlinear dynamics, and seasonality. A seasonal scale shared by two independent series is a separate

failure, specific to kernel tests. Without memory it drives HSIC with wild multipliers to 73.3%, while Pearson correlation stays at 4.7%. We prove that a shared seasonal mean with a random phase drives the size of the Pearson, wild, and subsampling rules to one. The same experiments show that circular shifts hold size under both seasonal forms, at every bandwidth we tried, and keep their power at matched size. These results yield the decision rule. Third, we apply the rule to separated Nasdaq trade-duration windows. There the null cannot be verified, so the rejection fractions are descriptive, not false-alarm rates. A smooth intraday adjustment does not remove the scale drift inside the stock windows, which matches the slow-drift limit of the rule. The two headline rates come from different nulls and different rules. They illustrate the problem and do not rank MMD against HSIC.

Long memory means that correlation fades slowly with time. Even a long series may therefore contain much less information than the same number of independent observations. Kernel features inherit some of this persistence. Their averages can vary more than a critical-value rule based on short windows or short multiplier correlations expects. We vary memory, transformations, sample size, and rule to see where this mismatch appears.

The two tests require separate explanations. Split MMD compares the first and second halves of one stationary series; its null says the halves have the same distribution. HSIC and distance covariance compare two aligned series; their null says the two processes are independent, although each may have memory within itself. For a rank-one split-MMD feature, the mean can change rate as soon as $H > 1/2$. For the product of two independent rank-one features, the covariance calculation instead points to $H_X + H_Y = 3/2$. Neither calculation provides a critical value for the exact kernel tests used here.

The rule matters as much as the statistic. On the same independent-fGn cell, $(H_X, H_Y) = (0.7, 0.7)$ and $n = 2000$, HSIC rejects 9.2% with subsampling, 7.35% with wild multipliers, and 5.45% with circular shifts. The wild rule also rejects 6.5% at the $H_X = H_Y = 0.5$ baseline. Table 3 separates that baseline error from the change associated with stronger memory.

Memory is not the only process feature that can break these tests. We therefore switch on one feature at a time and cross it with memory. Heavy tails and nonlinear dynamics change the size of the memory effect but do not create a large excess on their own. Seasonality does. A seasonal mean shared by two independent series misleads every test we ran, including Pearson correlation. This is the classical nonsense-correlation problem [Yule, 1926]. A seasonal *scale* behaves differently. With no memory, it leaves Pearson at 4.7% but drives HSIC with wild multipliers to 73.3%. The kernel test sees that the two series swing widely at the same times, even though their signs are unrelated. This sensitivity to more than linear correlation is the reason to use a kernel test. On ordered data it is also an extra exposure.

The final application uses AAPL, MSFT, and NVDA trade durations to test the boundary of this account in market microstructure. Paired windows are at least 20 trading days apart, but that gap does not prove process independence. We treat their rejection fractions as descriptive observations, not measured false-alarm rates.

Box 1. Practical guidance for independence tests on persistent series.

1. **Remove the seasonal mean and the seasonal scale from each series separately.** Removing the mean alone is not enough. Under a seasonal scale it leaves HSIC with wild multipliers at 73.8%, against 7.2% after scale adjustment (Section 4.5). A periodogram estimate of the period worked as well as the true period. For intraday data, the standard tool is a smooth diurnal factor [Engle and Russell, 1998, Wu, 2012]. Adjust before testing, because no shift can break a pattern that both series share in fixed phase. In that diagnostic, circular shifts reject 28%–34%.
2. **Test with HSIC and circular shifts.** At $n = 2000$, this rule rejected 3.8%–7.0% of true nulls in every simulated arm except two. These are the slow drift in item 3 and the shared fixed phase in item 1. Subsampling reached 19.7% at $H_X = H_Y = 0.8$, and wild multipliers reached 73.3% under a seasonal scale. At matched size, shifts stayed within 4.9 points of the best other HSIC rule in all 18 power cells (Section 4.4). Shifts also held size at every bandwidth from half to four times the median.
3. **Expect residual risk from slow drift.** A seasonal scale with period $4n$ shows each window only a quarter cycle, so adjustment cannot remove it. Without memory it raised circular shifts to 10.1% at $n = 2000$, against 6.2% at $n = 500$. We have no correction for this case. Treat a rejection as weak evidence when a window covers less than one cycle of a known pattern.
4. **Know what the rule does not promise.** No long-memory theory covers circular shifts. Under strong memory their rate creeps up with n. At $H_X + H_Y = 1.6$ it is 5.7% at $n = 2000$, 6.3% at $n = 4000$, and 8.3% at $n = 8000$ from 144 replications. Adjustment also costs size under memory. At $H = 0.8$ it raised circular shifts from 5.3% to 6.8%.
5. **For split MMD, no tested rule held size under memory.** Self-normalized blocks came closest, at 6.0%–8.5% on fGn. Memory also makes a fixed mean shift harder to see, because the half-means spread apart under the null (Proposition 2). This is a loss of information, not a kernel flaw. At a shift scaled to the null spread, MMD keeps its power under memory. Read a split-MMD rejection on a persistent series as a prompt to look further, not as evidence of change.

2 Theory and predictions

2.1 Memory and rank

Let ξ_t be a stationary Gaussian series. We write its autocovariance at lag k as

$$\rho(k) \sim ck^{-D}L(k), \qquad D = 2 - 2H, \quad 1/2 < H < 1, \tag{1}$$

where L varies slowly. The reference $H = 1/2$ has no long memory; larger H gives slower decay. Hermite rank m is the order of the first nonzero term when a centred feature $g(\xi_t)$ is expanded in Hermite polynomials. Its leading covariance has order $\rho(k)^m$. If $mD > 1$, that leading term is summable and suggests the usual square-root-n rate for a scalar partial sum. At equality, L can change the answer.

This calculation predicts different rates for different features. Rank one remains non-summable for every $H > 1/2$. Rank two reaches the balance $2D = 1$ at $H = 3/4$. It is a statement about transformed Gaussian averages, not a null law or critical value for MMD or HSIC. Long-memory two-sample U-process theory provides related results [Dehling et al., 2017].

A separate rank-two result for a Gaussian graphical-model statistic appears in Zhai et al. [2026]; its assumptions and statistic differ from ours.

2.2 Two-sample MMD

MMD asks whether two samples have the same distribution. Here they are the first and second halves of one observed series $Z_1, \ldots, Z_n$. We compare their kernel mean embeddings:

$$\mathrm{MMD}_n^2 = \left\| \frac{2}{n} \sum_{t \le n/2} \phi(Z_t) - \frac{2}{n} \sum_{t > n/2} \phi(Z_t) \right\|_{\mathcal{H}}^2 . \tag{2}$$

Stationarity makes the two population embeddings equal, so rejection under this generator is a false alarm. If the feature $\phi(G(\xi_t))$ has a nonzero rank-one term, its sample mean carries long memory for any $H > 1/2$. An even transform such as $G(\xi) = \xi^2 - 1$ can cancel that term and leave rank two as the leader. This is why we compare rank-one and even transforms. It predicts a difference in scale; it does not prove the null law of the implemented MMD statistic [Dehling et al., 2017].

2.3 Independence statistics

HSIC and distance covariance ask a different question: are the processes X and Y independent? Each process can still depend strongly on its own past. The leading cross-series term has the schematic form

$$U_t = \{f(X_t) - \mathbb{E}f(X_t)\}\{g(Y_t) - \mathbb{E}g(Y_t)\}. \tag{3}$$

If the two processes are independent and both factors have rank one, $\mathrm{Cov}(U_0, U_k)$ is the product of their within-series covariances. It decays at order $k^{-(D_X + D_Y)}$. The power-law balance lies at

$$D_X + D_Y = 1 \quad \Longleftrightarrow \quad H_X + H_Y = 3/2. \tag{4}$$

With equal memories, this corresponds to $H = 3/4$ in each series. We include unequal-memory pairs to see whether the sum organizes their rejection rates. The line is a covariance balance, not a universal point at which tests start to fail.

The closest published limit concerns a related statistic. Betken and Dehling [2026] prove a limit for the empirical joint characteristic-function process under independent Gaussian-subordinated series, with a distance-covariance corollary. On one side, their Gaussian conditions require *both* $D_X, D_Y > 1/2$, or both H values below $3/4$ [Betken and Dehling, 2026, Theorem 2.2(i) and Corollary 2.1(i)]. Their non-central result for $D_X + D_Y < 1$ uses Gaussian identity transforms [Betken and Dehling, 2026, Theorem 2.2(ii) and Corollary 2.1(ii)].

Those assumptions leave a mixed pair with one $H > 3/4$ but $H_X + H_Y < 3/2$ outside the stated result. The theorem also does not give a Gaussian-kernel HSIC limit. We therefore use Equation (4) to organize experiments, while keeping the proved distance-covariance case separate from the HSIC prediction.

Table 1: Scope of the theory used to motivate the experiments. No cited result validates every implemented critical-value rule.

Result	What it covers	What remains open here
Hermite covariance calculation	Rate of a scalar transformed Gaussian mean or independent product feature	RKHS-valued split-MMD and HSIC limits; critical-value laws
Betken and Dehling [2026], Thm. 2.2 and Cor. 2.1	Characteristic-function limit; distance-covariance corollary	Gaussian-kernel HSIC; mixed normal region; exact finite-sample rule below
Betken and Dehling [2026], Thm. 2.4	Subsampling for a statistic with a limit, under process and block conditions	Exact HSIC and dCov transfer; finite-sample size
Chwialkowski et al. [2014]	Wild bootstrap under weak-dependence assumptions	Long-memory validity of the present multiplier rule
Proposition 1	Random-phase seasonal mean: Pearson, wild, and subsampling sizes tend to one	Exact size of circular shifts (sketch only)
Proposition 2	Exact half-mean spread and a power cap for location-invariant split tests	Gaussian-kernel MMD power below the cap

Published critical-value results have their own entry conditions. The random-process HSIC test of Chwialkowski and Gretton [2014] and the distance-covariance block bootstrap of Betken et al. [2024] assume absolute regularity (β-mixing) at a polynomial rate. The RKHS self-normalized tests of Ghoshal and Shao [2026] assume summable strong-mixing coefficients; the authors note that this "requires the underlying temporal dependence to be of short-range type". Fractional Gaussian noise with $H > 1/2$ is not completely regular [Betken and Wendler, 2018, Example 1]. For a stationary Gaussian sequence, the maximal correlation is bounded by 2π times the strong-mixing coefficient [Bradley, 2005, Section 7], so such fGn is not strongly mixing and these conditions exclude our long-memory cells. The wild bootstrap of Chwialkowski et al. [2014] uses a τ-dependence coefficient. Taking one future observation and the identity as the one-Lipschitz test function gives $\tau(r) \geq \mathbb{E}|\mathbb{E}(Z_r \mid \mathcal{F}_0)|$. For a stationary Gaussian series, conditioning on the whole past leaves the conditional mean with variance at least $\rho(r)^2$, so $\tau(r) \geq (2/\pi)^{1/2}|\rho(r)|$. For fGn this bound decays like r^{2H-2}. Their rate conditions, $\sum_r r^2\tau(r)^{1/2} < \infty$ for the degenerate V-statistic limit and $\tau(r) = O(r^{-6-\epsilon})$ for MMD, therefore fail for every $H > 1/2$. Their MMD result also requires the two samples to be independent, which the two halves of one long-memory series are not. The HSIC wild bootstrap of Diz-Castro et al. [2026] assumes near-epoch dependence on a β-mixing process with summable rates. Its Gaussian limit has a covariance operator defined by a series over lags. For independent fGn pairs and a rank-one product feature, that series diverges when $H_X + H_Y \geq 3/2$ by Equation (4), so their conditions cannot hold there. Below that line, representing fGn through its own Gaussian innovations fails their rate condition, but we have not excluded every possible representation. Betken and Dehling [2026, Theorem 2.4] prove a subsampling result when a statistic has a nondegenerate limit and the process and block lengths meet their conditions. The theorem allows the offset $d_n = \lfloor 0.1n \rfloor$. Together with their distance-covariance limit, it may cover some independent-fGn dCov cases, but only after checking our exact statistic and block rule.

It does not validate our HSIC rule. An HSIC null limit under these assumptions is missing, and our subsampling windows reuse bandwidths selected from the full sample. Other long-memory subsampling and self-normalized block results also require statistic-specific conditions [Betken and Wendler, 2018, Bai et al., 2016]. Block choice can matter under Gaussian subordination [Zhang et al., 2022]. Table 1 separates these published results from what remains open here.

2.4 Two exact results

Two of the empirical findings follow from exact calculations. Appendix A gives the proofs. The first explains why a seasonal mean with a random phase misleads single-path tests. The second explains why memory removes the power of split MMD against a mean shift.

Throughout, ξ and ζ are independent unit-variance fGn sequences with Hurst parameter $H \in (0,1)$. A *single-path test* is a test whose decision is a measurable function of one observed pair $(X_t, Y_t)_{t=1}^n$, possibly with auxiliary randomness such as multiplier draws. All its reference values are computed from that same pair. The size of such a test is its rejection probability under the joint law of the phases and the noise. This is the quantity that EXP2 estimates.

Seasonal mean with a random phase. Fix an amplitude $a > 0$ and an integer period $P \geq 3$. Let U and V be independent and uniform on $\{0, \dots, P-1\}$, and independent of (ξ, ζ). Set

$$X_t = \xi_t + a \sin\{2\pi(t+U)/P\}, \qquad Y_t = \zeta_t + a \sin\{2\pi(t+V)/P\}. \tag{5}$$

The processes X and Y are independent. EXP2 uses $a = 1$ and $P = 50$. Let $D = (U - V) \bmod P$ be the phase gap. It is uniform on $\{0, \dots, P-1\}$. Define

$$c(d) = \frac{(a^2/2)\cos(2\pi d/P)}{1 + a^2/2}. \tag{6}$$

Let r_n be the sample Pearson correlation and $\hat{C}_n = n^{-1}\sum_t (X_t - \bar{X})(Y_t - \bar{Y})$ the sample covariance.

Proposition 1 (Seasonal mean)**.** *Under (5), the following hold.*

(i) $r_n \to c(D)$ *almost surely. The limit is random and has mean zero. Moreover* $\Pr\{c(D) = 0\} = 2/P$ *if* 4 *divides* P*, and* 0 *otherwise.*

(ii) Consider any test that rejects when $|\hat{C}_n| > z_n(\hat{v}_n/n)^{1/2}$*, where* z_n *is bounded and* $\hat{v}_n \geq 0$ *satisfies* $\hat{v}_n/n \to 0$ *in probability. Its size satisfies* $\liminf_n \Pr(\text{reject}) \geq \Pr\{c(D) \neq 0\}$*, which equals* 1 *for* $P = 50$*. The Newey–West Bartlett rule with bandwidth* $S_n = o_p(n)$ *and the equal-weighted cosine rule with* $\nu_n \to \infty$ *projections belong to this class. So do the implementations used here.*

(iii) Let $T_n = nV_n$ *be the Gaussian-kernel HSIC statistic of Section 3, with median-distance bandwidths, and let* $P \geq 5$*. Then* $V_n \to h(D)$ *almost surely for a function* h *with* $\min_d h(d) > 0$*. Any test that rejects when* $T_n > \kappa_n$*, with* $\kappa_n/n \to 0$ *in probability, therefore has size tending to* 1*. The window-subsampling rule with window length* $\ell_n = o(n)$ *belongs to this class. The block-Markov wild rule with* $\ell_n = o(n)$ *and a fixed number* $B \geq \alpha^{-1} - 1$ *of draws, centred or raw, also has size tending to* 1*.*

The proposition does not say that every single-path test fails. It covers tests whose critical value is of smaller order than the statistic's seasonal component. For Pearson that means a critical value for r_n that tends to zero. For HSIC it means a critical value for T_n of order $o(n)$. The circular-shift rule is not in this class, because its reference values are themselves of order n (Remark 1).

The limit also explains the finite-sample rate. With $a = 1$ and $P = 50$, the gaps $d \in \{12, 13, 37, 38\}$ give $|c(d)| = 0.021$, and every other gap gives $|c(d)| \geq 0.06$. At $n = 2000$, $\sqrt{n}\,|c(d)| \approx 0.9$ at those four gaps, so a 5% test rarely rejects there. Those gaps carry probability $4/50$, which predicts a rejection rate near 92%. EXP2 records 91.4% for Pearson NW. The HSIC limit $h(d)$ is also smallest at these gaps.

Split-sample mean shift. Let $n = 2m$ and $Z_t = \mu + \xi_t + \delta\,\mathbf{1}\{t > m\}$, with μ unknown. Write Γ for the $n \times n$ fGn covariance matrix, $\mathbf{1}$ for the vector of ones, and $c = (0, \dots, 0, 1, \dots, 1)^\top$ for the second-half indicator. Let z_q be the standard normal q-quantile and Φ its distribution function.

Proposition 2 (Half-mean spread). *(i) Let* $\Delta_m = m^{-1}\sum_{t\le m}\xi_t - m^{-1}\sum_{t>m}\xi_t$. *Then, exactly for every* m,

$$\mathrm{Var}(\Delta_m) = \frac{4m^{2H} - (2m)^{2H}}{m^2} = (4 - 2^{2H})\, m^{2H-2}. \tag{7}$$

Hence the half-mean contrast has signal-to-spread ratio $R_m = \delta\, m^{1-H}/(4-2^{2H})^{1/2}$.

(ii) The two-sided level-α *test that rejects when the observed half-mean difference* $|\bar{Z}_{1:m} - \bar{Z}_{m+1:2m}| = |\Delta_m - \delta|$ *exceeds* $z_{1-\alpha/2}\,\mathrm{Var}(\Delta_m)^{1/2}$ *has power* $\Phi(R_m - z_{1-\alpha/2}) + \Phi(-R_m - z_{1-\alpha/2})$.

(iii) Let ϕ *be any test, possibly randomized, that is invariant under* $Z \mapsto Z + b\mathbf{1}$ *for every real* b, *and has level* α *when* $\delta = 0$. *Then its power at* $\delta > 0$ *is at most* $1 - \Phi(z_{1-\alpha} - \delta I_n^{1/2})$, *where*

$$I_n = c^\top \Gamma^{-1} c - \frac{(\mathbf{1}^\top \Gamma^{-1} c)^2}{\mathbf{1}^\top \Gamma^{-1}\mathbf{1}} \tag{8}$$

is the inverse variance of the generalized-least-squares estimator of δ. *The matrix* Γ *is invertible because the fGn spectral density is positive.*

At $n = 2000$ and $\delta = 0.1$, part (i) gives $R_m = 2.24$ at $H = 0.5$, 0.68 at $H = 0.7$, and 0.40 at $H = 0.8$. Part (ii) then gives a power of 61%, 10.5%, and 7% for the linear half-mean test at 5%. A direct Toeplitz solve gives $\delta I_n^{1/2} = 2.24$, 0.71, and 0.46. Part (iii) therefore caps the power of every location-invariant 5% test at 72%, 18%, and 12%. Gaussian-kernel split MMD is location-invariant. Its kernel and median bandwidth depend on the data only through differences $Z_s - Z_t$. The wild, subsampling, and self-normalized rules are built from the same kernel values, and the size-adjusted cutoff is a constant. Every MMD rule in Table 5 is therefore bound by part (iii), up to Monte Carlo error in the size adjustment. The observed 6.4–7.5% at $H = 0.8$ lies under the 12% cap, and the 7–9% at $H = 0.7$ lies under the 18% cap. The loss of power is a loss of information in the data, not a failure of a kernel rule.

Part (iii) is a bound, not a description of the Gaussian kernel. A heuristic links the two. The rank-one Hermite coefficient of the feature map is $a_1 = \mathbb{E}\{\xi_0\phi(\xi_0)\}$. By Stein's identity it equals the derivative of the mean embedding of $\xi_0 + \delta$ at $\delta = 0$. To first order in δ, the shift therefore moves the embedding along a_1, and the rank-one noise along a_1 is $\Delta_m a_1$. Along this direction the signal-to-spread ratio is R_m. A Hermite term of rank r adds noise of variance order $m^{\max\{r(2H-2),-1\}}$, up to a logarithm at equality, in other directions. At $H = 0.5$ they have the same order as the rank-one term, which may explain why oracle MMD (45.7%) falls short of the linear test (61%). At $H = 0.8$ the rank-one term dominates and the two agree (6.4% against 7%). We have not proved this expansion for the RKHS-valued statistic.

3 Experiments and critical-value rules

Every rule targets a nominal 5% rejection level. In a simulation cell, we know that the null is true: the independence arms generate independent processes, and the MMD arm splits one stationary process. Its rejection fraction is therefore an empirical false-alarm rate. We do not know whether the process-independence null holds for stock trades, so their fractions are *placebo rejection rates*. Wilson intervals measure counting uncertainty for simulation fractions, not uncertainty about the generator or the rule.

Table 2: Study arms. A simulation cell's null is fixed by its generator; the stock placebo has an unverified process-independence null.

Arm	Series and comparison	Sample size	Replications or pairs
Initial MMD	Stationary fGn and transforms, split in halves	500, 1,000, 2,000, 4,000	1,000–2,000
Initial independence	Independent fGn, subordinated and unequal-memory pairs	500, 1,000, 2,000	2,000 per cell
EXP1 H5/E2	MMD rule control and rank-one/two transforms	500, 1,000, 2,000	1,000–2,000
EXP1 E3	Ten equal or unequal independent fGn pairs	500, 1,000, 2,000	2,000 per cell
EXP3	Size-adjusted power: linear, squared, and seasonal-scale dependence, and a split-MMD mean shift	500, 2,000	1,000 per cell
EXP4	EXP2 seasonal and Gaussian nulls after estimated seasonal adjustment, and a slow scale drift	500, 2,000	1,000 per cell
EXP5	Split-MMD power at a scaled shift, and independence nulls at larger n	500–8,000	144–1,000 per cell
EXP6	HSIC size and power at four bandwidth multiples	2,000	500 per cell
EXP2	Gaussian, heavy-tailed, nonlinear, and seasonal generators crossed with $H \in \{0.5, 0.7, 0.8\}$	500, 2,000	1,000 per cell
Stock placebo	Separated AAPL, MSFT, NVDA duration windows	500, 1,000, 2,000	500 pairs per cell
Revision control	Independent $H = 0.5$ fGn pairs; centred versus raw multipliers	500, 1,000, 2,000	1,000 per cell

We also compared centred and raw Markov multipliers on independent $H = 0.5$ Gaussian pairs. This control measures rule error before long memory is introduced. Its counts and selection rule are in Appendix B.

3.1 Initial MMD and independence experiments

The first MMD experiment splits each series in half. It uses fGn, the bounded transform $2\Phi(\xi)-1$, and $\operatorname{sign}(\xi)$. The fGn grid is $H \in \{0.5, 0.6, 0.7, 0.75, 0.8, 0.9\}$. We compare wild-multiplier and sampling-window critical-value rules.

We include a slow-mean-reversion comparison to distinguish persistent behaviour from a different dependence pattern. It follows $Y_t = (1-\lambda)Y_{t-1} + \varepsilon_t$, with fGn innovations at $H = 0.1$, $\lambda \in \{0.01, 0.1\}$, and a burn-in of $\lceil 10/\lambda \rceil$ observations. Generated copies are independent of one another, although innovations within a copy are correlated. This is a discrete mean-reverting model, not a claim of continuous-time fractional OU sampling.

The independence experiment compares Gaussian-kernel HSIC, distance covariance, and Pearson correlation as a linear reference. It uses independent fGn pairs, bounded and sign transforms, unequal-memory pairs, and independent copies of the mean-reverting process. Kernel sample sizes are $n \in \{500, 1000, 2000\}$; the Pearson reference uses $T \in \{500, 2000, 8000\}$.

The rules construct critical values in different ways. For HSIC and distance covariance, the wild rule uses 500 draws of empirically centred block-Markov Rademacher multipliers. Their correlation decays as $\exp(-k/\ell)$, with $\ell = \lfloor \sqrt{n} \rfloor$. The HSIC shift rule realigns one full series by circular offsets from $\lfloor 0.1n \rfloor$ to $n - \lfloor 0.1n \rfloor$. Subsampling instead measures contiguous windows of length $\ell = \lfloor \sqrt{n} \rfloor$, with a cross-series offset of $\lfloor 0.1n \rfloor$. Pearson uses Newey–West (NW)

and equal-weighted-cosine (EWC) long-run variance rules. These settings describe the tested implementations; they are not general validity claims.

Here is the exact statistic used for HSIC. Let K^c and L^c be empirically double-centred Gaussian Gram matrices. The bandwidth for each is the median off-diagonal pairwise distance, with a positive-distance fallback for ties. We compute $T_n = n^{-1}\sum_{i,j} K^c_{ij}L^c_{ij}$. Distance covariance uses double-centred distance matrices in the same construction. In either case, $T_n = nV_n$ for the corresponding biased V-statistic.

For one multiplier draw w, the HSIC reference value is $T^*_n = n^{-1}w^\top(K^c \circ L^c)w$; distance covariance uses the analogous distance product. With B reference draws, the upper-tail p-value is $(1+\#\{b : T^*_b \geq T_n\})/(B+1)$. The shift rule applies the same counting convention to zero-shift HSIC and the permitted circular alignments.

For subsampling, the full-series value is $T_n = nV_n$. Each window has its own value $T_\ell = \ell V_\ell$, and the rule rejects when T_n exceeds the empirical 95th percentile of the window values. The fGn generator is Davies–Harte. This specification fixes what was computed; it does not establish the rule's asymptotic validity.

3.2 Rule checks, rank change, and finer memory grid

The simulation extension has three parts. H5 compares five MMD multiplier variants at $H = 0.5$ to find rule error when long memory is absent. E2 tests the even rank-two transforms $\xi^2 - 1$ and $|\xi|$, adds $H = 0.55$, and compares multiplier lengths, two subsampling block rules, plug-in rate correction, and self-normalized MMD. E3 tests independence across a finer memory-sum grid.

Self-normalized MMD divides $n\,\mathrm{MMD}^2_n$ by a within-series measure of variation: $T^{\mathrm{SN}}_n = n\,\mathrm{MMD}^2_n\,/V_n$. The denominator V_n sums squared deviations of RKHS partial sums from straight-line bridges within each half, then divides by n^2. The critical value is the 95th percentile of this same ratio on every contiguous window of length $\lfloor n^{0.5}\rfloor$ or $\lfloor n^{0.7}\rfloor$. Under suitable conditions, the ratio can remove an unknown common scale. We do not assume those conditions validate this rule here. Appendix B gives the exact V_n.

E3 uses ten equal and unequal independent fGn pairs with $H_X + H_Y$ from 1.3 to 1.7. Each $n \in \{500, 1000, 2000\}$ cell has 2,000 replications. We plot the rates and changes between adjacent grid points; we do not fit a breakpoint.

3.3 Process features crossed with memory

EXP2 asks whether a test fails because of memory alone, a process feature alone, or the two together. Let ξ_t be unit-variance fGn with Hurst parameter H. At $H = 0.5$ it is independent standard Gaussian noise. Every generator is driven by this ξ_t, so the Gaussian row shares the same code path. We switch on one feature at a time:

- *G0, Gaussian baseline:* $X_t = \xi_t$.
- *HT-S, instantaneous heavy tails:* $X_t = F^{-1}_{t_3}\{\Phi(\xi_t)\}/\sqrt{3}$. This is an odd, increasing function of ξ_t, so it has Hermite rank one and the rank calculation of Section 2 applies.
- *HT-L, filtered heavy tails:* an ARFIMA$(0, H - 1/2, 0)$ filter of independent t_3 innovations. This process is not a function of a Gaussian series.
- *NL-SV, stochastic volatility:* $X_t = \exp(h_t/2)\varepsilon_t$, where h_t is a unit-variance AR(1) filter of ξ_t with coefficient 0.9 and ε_t is independent Gaussian noise. The series is uncorrelated, and its memory sits in its volatility.
- *NL-TAR, threshold autoregression:* $X_t = a(X_{t-1})X_{t-1} + \xi_t$, with $a(x) = 0.8$ for $x \leq 0$ and -0.4 otherwise.
- *SEAS-M, seasonal mean:* $X_t = \xi_t + \sin\{2\pi(t + U)/50\}$.
- *SEAS-V, seasonal scale:* $X_t = [1 + 0.5\sin\{2\pi(t + U)/50\}]\xi_t$.

The phase U is uniform on $\{0, \ldots, 49\}$ and is drawn afresh for each series. Each series in an independence pair therefore has its own phase, and the null of independent processes holds

exactly. A diagnostic row gives both series the same fixed phase. These settings are illustrative and are not calibrated to market data.

Each generator is crossed with $H \in \{0.5, 0.7, 0.8\}$ and $n \in \{500, 2000\}$, with 1,000 replications per cell. The independence arm uses the HSIC, distance-covariance, and Pearson rules described above. The MMD arm uses the uncentred wild rule, square-root subsampling, and self-normalized blocks. We read two effects. The *alone* effect is the rate at $H = 0.5$ minus the Gaussian rate at $H = 0.5$. The *interaction* asks whether the rise from $H = 0.5$ to $H = 0.7$, and from $H = 0.5$ to $H = 0.8$, differs from the Gaussian rise. The hypotheses, generator checks, and a pilot were fixed in a design record before the main run.

3.4 Size-adjusted power and seasonal adjustment

EXP3 measures power at a matched size. Each rule is reduced to one continuous number per replication, the ratio of its statistic to its own 95% reference quantile. The raw rule rejects when this ratio exceeds one. The size-adjusted rule rejects when it exceeds the 95% quantile of the same ratio over 1,000 replications of the matched null cell, so its null rejection rate is 5% by construction. This adjustment uses the true null of a simulated cell, which an analyst never has. It shows whether a statistic and rule carry information, not how to test real data. We also size-adjust the bare statistic, as an oracle that shows what each critical-value rule costs. Let ξ and w be independent fGn series. The independence alternatives are linear, $Y = \rho\xi + (1-\rho^2)^{1/2}w$, and squared, $Y = \rho(\xi^2 - 1)/\sqrt{2} + (1-\rho^2)^{1/2}w$, with $X = \xi$. A third alternative applies independent seasonal scales to the linear pair. The MMD alternative adds δ to the second half of one series. At $n = 2000$, $\rho = 0.065$ for the linear pairs, $\rho = 0.075$ for the squared pair, and $\delta = 0.1$. At $n = 500$ each effect is doubled. Intervals come from a bootstrap that resamples both the null and the alternative replications.

EXP4 adjusts each series of a pair separately and then reruns the EXP2 independence rules. Adjustment A1 subtracts period-bin means, with bin t mod 50. A2 also divides by period-bin standard deviations. A3 estimates the period from the periodogram and then applies A2. Every adjustment uses only the series it adjusts. A further diagnostic, SEAS-V-slow, gives each series a seasonal scale with period $4n$, so that one window sees about a quarter cycle. This mimics the stock arm, where a window of 2,000 durations covers a few minutes of a trading day, and no per-window bin adjustment is possible. Every EXP4 cell has 1,000 replications.

Two follow-up studies test the robustness of these results. EXP5 repeats the MMD power study with the shift scaled to the null spread of the half-mean difference, and it runs the equal-memory independence nulls at $n = 4000$ and, with 144 replications, $n = 8000$. EXP6 multiplies the median bandwidth by $\kappa \in \{0.5, 1, 2, 4\}$ in the HSIC size and power cells at $n = 2000$, with 500 replications per cell. At $\kappa = 1$ it reproduces 30 saved EXP3 replications exactly.

3.5 Stock-duration placebo

The applied arm uses Databento XNAS.ITCH trade records[1] for AAPL, MSFT, and NVDA from January through June 2024. We keep trades from 09:30 to 16:00 New York time. Within each day, we sort by event timestamp and feed sequence, collapse prints with the same event timestamp, and measure the positive gaps between distinct timestamps. A gap is one intertrade duration.

We take the log of each duration in seconds and subtract that stock's pooled mean for its 15-minute clock bin. This removes a coarse time-of-day pattern. The local cache also keeps the raw gap, timestamp, day, and clock bin for checks described in Appendix B. As a control, we rerun the same windows and seeds under two smooth adjustments, each fitted per stock on the pooled sample. D1 divides each duration by a spline estimate of its expected value at that

[1] Dataset and trade-schema documentation: https://databento.com/datasets/XNAS.ITCH and https://databento.com/docs/schemas/trades.

time of day, the multiplicative diurnal factor of Engle and Russell [1998], and takes logs. D2 also removes a spline estimate of the diurnal mean and standard deviation of the D1 values. Removing each day's mean as well would change no test, because every window lies inside one day.

Each placebo pair contains two random-start windows of 500, 1,000, or 2,000 consecutive durations. A window stays within one day; the paired days are at least 20 trading sessions apart. We test AAPL–MSFT, AAPL–NVDA, MSFT–NVDA, and AAPL–AAPL, with 500 pairs per cell and 12 cells in total. The same-stock pair checks the sampling construction; it does not establish independence. We reuse the simulation's HSIC, distance-covariance, and Pearson implementations. Days can recur across pairs, and the time-of-day adjustment uses pooled data. These facts prevent us from treating the observed rejection fractions as known-null false-alarm rates.

4 Results

Table 3 gives the main comparison in one place. The $H_X = H_Y = 0.5$ column shows each rule's baseline error. The next two columns show rates as within-series memory strengthens. The change from $n = 500$ to $n = 2000$ shows that excess rejection does not necessarily grow over the sample sizes we tested.

Table 3: Independent-fGn false-alarm rates at nominal 5%. The first three numeric columns are percentages at $n = 2000$ for equal-memory pairs with $(H_X, H_Y) = (0.5, 0.5), (0.7, 0.7), (0.8, 0.8)$. Δ is the percentage-point change from $n = 500$ to $n = 2000$ at sum 1.6. Each cell has 2,000 replications. Theory status refers to the exact rule used here. At $n = 4000$ every rule rejects more than at $n = 2000$ at sum 1.6 (Section 4), so Δ is not a trend with n.

Rule	$H = .5$	Sum 1.4	Sum 1.6	Δ	What it limits	Theory status
HSIC wild	6.5	7.4	8.9	−2.4	Downweights lags on a $\sqrt{n}$ scale	Weak-dependence theory only
HSIC shift	5.0	5.5	5.7	−0.3	Excludes shifts $< 0.1n$; uses full series	No long-memory proof here
HSIC sub	5.6	9.2	19.7	−2.2	Uses only $\sqrt{n}$ windows	HSIC limit and rule match open
dCov wild	6.6	7.1	9.6	−2.2	Downweights lags on a $\sqrt{n}$ scale	Weak-dependence theory only
dCov sub	5.3	10.6	23.0	−2.2	Uses only $\sqrt{n}$ windows	Restricted dCov limit; exact rule open
Pearson NW	5.8	8.2	13.3	−0.3	Drops lags beyond fitted cutoff	Standard HAC excludes these cells
Pearson EWC	6.1	6.0	8.5	+0.8	Drops modes beyond $\lfloor 0.4n^{2/3} \rfloor$	Long-memory guarantee open

4.1 Split MMD: rank changes when the false alarms rise

The first MMD experiment shows an early increase for rank-one features. At a nominal 5% level, its rejection rate is about 12–19% by $H = 0.6$ and 44–71% by $H = 0.9$, depending on the process and rule. This pattern agrees with the slow rank-one covariance decay. The tested grid does not locate a precise breakpoint.

We first checked the rule at the memory-free baseline. For Gaussian fGn at $H = 0.5$ and $n = 2000$, the old and published centred MMD multipliers reject 7.0% and 7.2% of true nulls. Independent and uncentred variants reject 5.3% and 5.0%. At $n = 500$, the old and centred rules reject about 9%. E2 therefore uses the uncentred rule that is closest to 5% in this control. This is a choice for the experiment, not a general claim that uncentred multipliers are better.

With that rule fixed, E2 compares feature ranks. At $n = 2000$, the even transform $\xi^2 - 1$ stays between 4.2% and 6.4% through $H = 0.75$, then rises to 11.3% at $H = 0.8$ and 22.8% at $H = 0.9$. The $|\xi|$ curve is similar. By contrast, the rank-one reference is already above nominal at $H = 0.55$ (Figure 1). Self-normalized block MMD varies less across the grid but still rejects about 6.0–8.5% on fGn and 4.7–7.5% on signs.

There is a cost to some apparent size improvements. In one $H = 0.7$ fGn shift alternative at $n = 2000$, self-normalized MMD detects the shift in 51% of replications. Conventional wild and subsampling rules detect it in 76–87% at their own cutoffs. Because their null rejection rates differ, this is not a size-adjusted power comparison. Section 4.4 gives one. Plug-in rate correction is often conservative and also detects fewer alternatives in this cell.

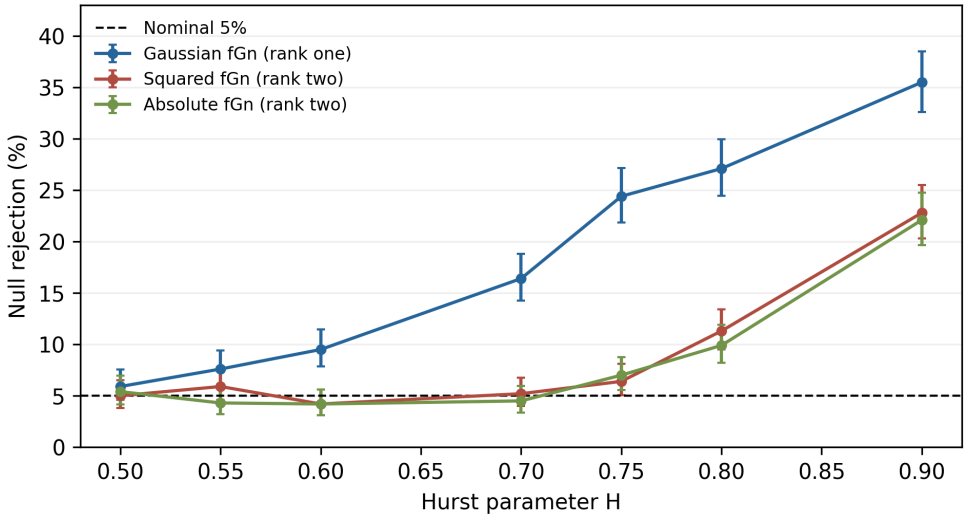


Figure 1: MMD rejection under the stationary split-series null at $n = 2000$, using the tested uncentred wild multiplier with length $\lfloor n^{0.5} \rfloor$. Points are saved E2 cell rates; bars are Wilson 95% counting intervals from 1,000 replications. The lines compare process-specific finite-sample behaviour, not proved asymptotic thresholds.

4.2 Independent pairs: the rule changes the answer

The independent-pair experiments show a different pattern. At $(H_X, H_Y) = (0.7, 0.7)$ and $n = 2000$, HSIC subsampling rejects 9.2% of true nulls and distance-covariance subsampling rejects 10.6%. On those same pairs, HSIC rejects 7.35% with wild multipliers and 5.45% with circular shifts. Pearson with NW and EWC references rejects 8.15% and 5.95%. Thus the critical-value rule changes the measured false-alarm rate even though the data and statistic are fixed. Oversized distance-covariance subsampling is not new. Betken and Dehling [2026, Appendix C, Table 1] report empirical sizes of 7.3–13.1% at $H = 0.6$ and 11.2–17.5% at $H = 0.7$ for a lag-weighted dCov statistic on independent fGn pairs, with $n \leq 1000$ and blocks of length $n^{0.4}$ or $n^{0.5}$. Our dCov subsampling rows extend that observation to $n = 2000$ and place it beside the wild and shift rules and the Gaussian-kernel HSIC statistic.

This excess appears below the covariance balance $H_X + H_Y = 1.5$. HSIC subsampling stays near 9.2% across $n = 500, 1000, 2000$ (9.1%, 9.15%, 9.2%). The wild-multiplier rule rejects 6.45% even at the $H_X = H_Y = 0.5$ baseline and $n = 2000$. Its excess at sum 1.4 therefore includes an error already present without long memory. At higher memory, subsampling becomes much more oversized, while circular shifts stay closest to 5% on this grid (Table 3). None of these patterns proves a rule valid outside the tested cells. EXP5 extends the equal-memory cells to $n = 4000$. At sum 1.4 the rates stay flat. HSIC subsampling rejects 8.2% and distance-covariance subsampling 9.4%. At sum 1.6 every rule rejects more at $n = 4000$ than at $n = 2000$. Distance-covariance

subsampling rises from 22.9% to 26.9%, and circular shifts from 5.7% to 6.3%. The small decline from $n = 500$ to $n = 2000$ in Table 3 therefore does not continue. A small run at $n = 8000$, with 144 replications per cell, points the same way.

The unequal pair $(0.55, 0.85)$ has rates similar to the equal pair $(0.7, 0.7)$. Both have memory sum 1.4, so the sum is useful for organizing this grid. Figure 2 shows the rates against that sum; Figure 3 shows changes between adjacent grid points. The algebraic line at 1.5 helps read the plots, but the plots do not locate a common rejection boundary or prove a mixed-memory limit for these tests.

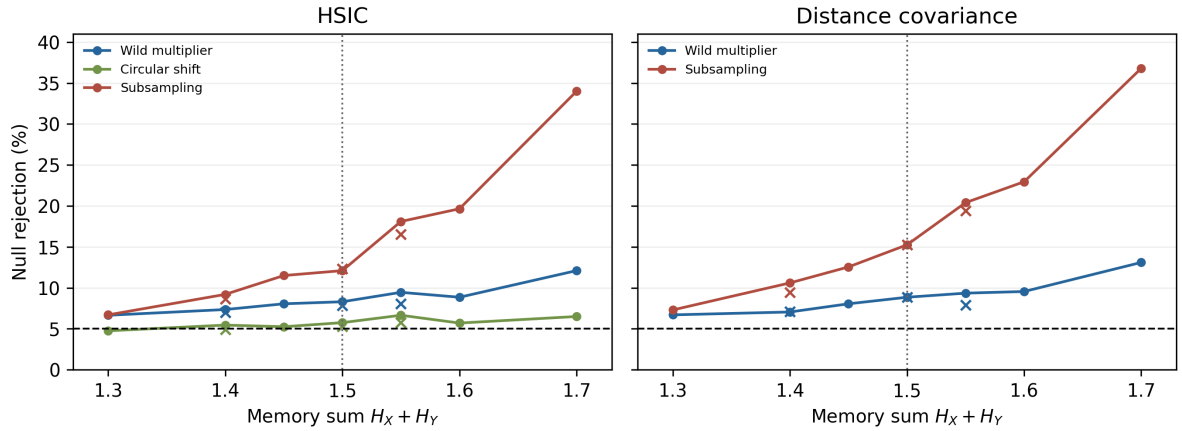


Figure 2: E3 null rejection at $n = 2000$ for independent fGn pairs. Connected circles are equal-memory pairs; crosses are unequal pairs. The horizontal dashed line is the nominal 5% rate and the vertical dotted line marks the candidate sum $H_X + H_Y = 1.5$. Each point has 2,000 replications. No line is a proved rejection boundary.

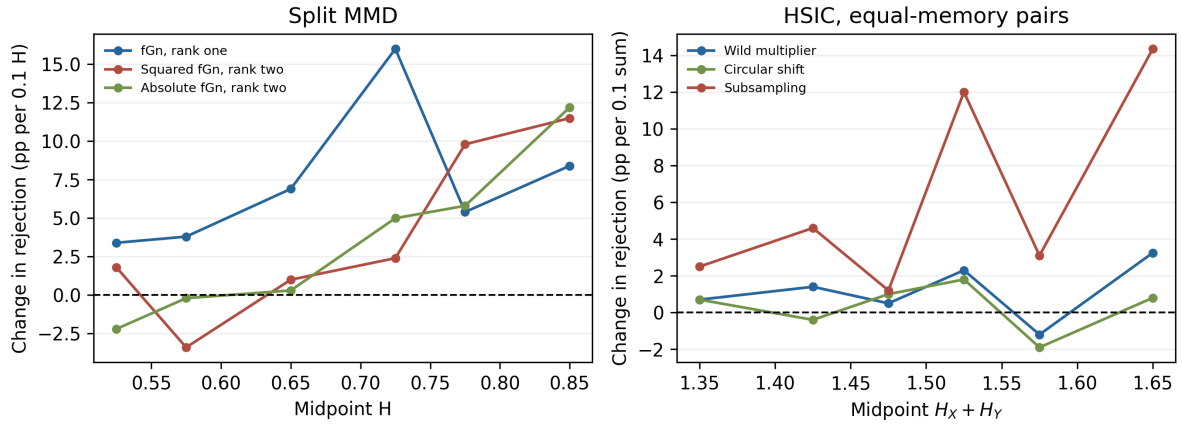


Figure 3: Adjacent-grid changes in null rejection at $n = 2000$. Left: split MMD under the uncentred wild rule, measured in percentage points per 0.1 increase in H. Right: HSIC on equal-memory independent pairs, measured per 0.1 increase in $H_X + H_Y$. Points sit at interval midpoints. These finite differences are descriptive, depend on grid spacing and Monte Carlo error, and do not estimate a change point.

We also tested whether raw multipliers would fix the memory-free HSIC and distance-covariance baseline. They moved rejection from above 5% to below it and did not improve every cell. We therefore kept the original centred rule for the stock placebo. Appendix B gives the paired counts and selection rule.

4.3 Process features: seasonal scale is a kernel-specific failure

Table 4 gives the independence results of EXP2 at $n = 2000$. Four findings stand out. Monte Carlo standard errors are about 0.7 points near 5% and 1.6 points near 50%. We correct the 288 tests behind the EXP2 verdicts with Holm's step-down method. These are 96 one-sided tests of an effect without memory and 192 two-sided tests of a change in the memory effect. Holm's method controls the familywise error under any dependence between the tests. Claims that compare rejection levels at $H = 0.8$ are corrected in a larger family, which adds 96 level comparisons for 384 tests in all. Every claim below survives this correction unless we say otherwise.

Table 4: EXP2 null rejection percentages for independent pairs at $n = 2000$ and nominal 5%, with $H_X = H_Y = H$. Each cell has 1,000 replications. The last row is a diagnostic in which both series share one fixed seasonal phase. It is excluded from the effect estimates.

	HSIC wild		HSIC shift		HSIC sub		dCov sub		Pearson NW	
Generator	.5	.8	.5	.8	.5	.8	.5	.8	.5	.8
G0 Gaussian	6.8	8.1	5.3	5.3	5.8	17.5	6.8	22.8	5.0	14.1
HT-S tails, instantaneous	6.1	7.0	4.5	3.8	4.3	15.7	3.2	16.5	5.7	11.5
HT-L tails, filtered	8.9	10.6	6.4	5.3	6.8	25.9	3.2	25.9	4.2	16.2
NL-SV volatility	7.9	6.7	5.8	3.9	4.9	6.2	2.2	1.9	5.2	4.6
NL-TAR threshold AR	7.1	11.9	4.7	7.0	7.2	29.8	6.4	33.5	6.6	12.1
SEAS-M seasonal mean	91.2	85.2	5.8	5.9	90.9	85.4	91.3	86.6	91.4	84.5
SEAS-V seasonal scale	73.3	68.0	5.0	5.1	71.5	73.0	45.6	53.3	4.7	13.4
Shared fixed phase	100.0	100.0	34.3	27.9	100.0	100.0	100.0	100.0	100.0	100.0

First, a seasonal mean misleads every single-path test, kernel or linear. With no memory, HSIC, distance covariance, and Pearson each reject about 91% of independent pairs. The phases are independent, so the processes are independent. Within one path, however, each series carries a fixed sinusoid with the same period. The sample correlation of two such sinusoids depends on their phase gap and does not shrink as n grows. The random phase acts as a component that never forgets, so a single path cannot average it out. This is the nonsense-correlation problem of Yule [1926], not a kernel-specific failure. Proposition 1 makes it exact. The sample correlation converges to a random nonzero limit, and the Pearson, wild, and subsampling rules used here have size tending to one. The limit predicts a Pearson rate near 92% at $n = 2000$, against the 91.4% observed. Circular shifts stay at 5.8%. A shift moves one series against the other and re-randomizes their phase gap, which is what the null requires. Remark 1 sketches why this keeps the size near 5%, but we have not proved it. Small declines at $H = 0.8$ in these rows reflect a ceiling near 90%, not a protective effect of memory.

Second, a seasonal scale misleads the kernel tests but not Pearson. With no memory, HSIC rejects 73.3% with wild multipliers and 71.5% with subsampling, and distance covariance rejects 45.6% with subsampling. Pearson-NW rejects 4.7%. A shared seasonal scale makes $|X_t|$ and $|Y_t|$ large at the same times while their signs stay unrelated. Pearson correlation sees only the linear layer, which is unaffected. HSIC and distance covariance also see the squared layer. SEAS-V is not a function of a Gaussian series, so the Hermite calculation of Section 2 does not apply directly. Informally, the shared pattern lives in the squared features of the two series, the analogue of the second Hermite layer that the even transforms isolate in the MMD experiment. Circular shifts again stay at 5.0%. When both series share one fixed phase, even circular shifts reject 28–34%, because no shift can break a common deterministic pattern.

Third, heavy tails matter little. The instantaneous t_3 transform tracks the Gaussian row within noise, as its rank-one structure predicts. The filtered t_3 process raises HSIC subsampling to 25.9% at $H = 0.8$, against 17.5% for the Gaussian row (Holm $p \leq 0.002$). Its gain over the Gaussian memory effect, 7.4 points, does not survive correction (Holm $p = 0.12$). For

distance-covariance subsampling, the filtered process is within noise of the Gaussian row at $H = 0.8$.

Fourth, the two nonlinear processes move the memory effect in opposite directions. Stochastic volatility places the memory in the variance of each series. HSIC and distance-covariance subsampling then stay at or below 6.2% even at $H = 0.8$, against 17.5% and 22.8% for the Gaussian row. The threshold autoregression adds 11–12 points to both subsampling rules at $H = 0.8$. That process also adds short-range persistence of its own. Its mean local-Whittle memory estimate is 0.59 at nominal $H = 0.5$ and $n = 2000$, against 0.50 for the Gaussian row. Its excess is therefore not a pure nonlinearity effect.

The MMD arm agrees with this picture. Uncentred wild MMD rejects 26–31% at $H = 0.8$ across the five non-seasonal generators, including 26.5% for the Gaussian row. Stochastic volatility breaks square-root subsampling even without memory (13.3% at $H = 0.5$), because its AR(0.9) volatility persists beyond a $\sqrt{n}$ window. Self-normalized blocks stay between 3.1% and 13.0% across the non-seasonal generators and memory levels. Seasonal series make split MMD conservative (0–2% at $H = 0.5$), because both halves contain whole seasonal cycles.

4.4 Power at matched size

Table 5 gives size-adjusted power at $n = 2000$. Circular-shift HSIC is competitive in all 18 HSIC cells. We call it competitive when its power is at most 5 points below the best other HSIC rule. Its largest shortfall is 4.9 points, at $n = 500$. It also matches the oracle HSIC within about 4 points. Its good size therefore does not come from rarely rejecting. The wild and subsampling rules show high raw power in several cells, but much of it is false alarm. Under the seasonal scale, for example, HSIC subsampling detects the linear alternative 86.4% of the time at its nominal cutoff, but only 14.6% at matched size.

Table 5: EXP3 size-adjusted power percentages at $n = 2000$, with each rule tuned to a true 5% null rate on its matched null cell. Each cell has 1,000 null and 1,000 alternative replications. The oracle row size-adjusts the bare HSIC or MMD statistic. Bootstrap intervals are about ±5 points.

	Linear		Squared		Seasonal scale	
Rule	$H = .5$	.8	.5	.8	.5	.8
HSIC wild	56.0	28.7	43.0	19.1	13.8	12.7
HSIC shift	57.2	30.6	44.0	21.2	15.2	17.4
HSIC sub	54.4	27.1	40.9	17.1	14.6	15.6
dCov sub	76.4	29.6	32.7	10.1	48.7	28.8
Pearson NW	82.9	36.9	4.4	8.6	76.4	34.6
Oracle HSIC	55.7	28.6	42.4	17.4	13.5	13.7
	Mean shift					
MMD unc. wild	47.1	7.5				
MMD sqrt sub	44.1	7.4				
MMD SN block	41.7	7.0				
Oracle MMD	45.7	6.4				

Three further patterns stand out. First, Pearson correlation is the most powerful test against linear dependence and has no power against the squared alternative, as expected. Kernel tests are the only ones that detect the squared alternative. Second, memory lowers power for every rule, because the same small dependence is harder to see in a persistent series. Third, memory removes almost all the power of split MMD against a fixed mean shift. At matched size, its detection rate falls from about 45% at $H = 0.5$ to 7–9% at $H = 0.7$ and 0.8, and the oracle falls the same way. The loss therefore lies in the statistic, not in a critical-value rule. Proposition 2 makes this exact. A shift of $\delta = 0.1$ is 2.24 null standard deviations of the half-mean difference at

$H = 0.5$ but 0.40 at $H = 0.8$. No location-invariant 5% test can exceed 12% power at $H = 0.8$, and every MMD rule here is location-invariant. EXP5 checks this by scaling the shift to the null spread, $\delta_H = 2.24\ \mathrm{Var}(\Delta_m)^{1/2}$, which equals 0.1 at $H = 0.5$. At matched size, MMD power then rises from 48% at $H = 0.5$ to 58–61% at $H = 0.8$ for the wild, subsampling, and oracle versions. The half-mean test moves from 66% to 59%. So the fixed-shift loss reflects the information in the data. Without memory MMD trails the half-mean test by about 18 points, and under strong memory the two are equal. Self-normalized blocks cost about 15 points at $H = 0.8$. Under the seasonal scale, HSIC loses most of its power at matched size, while distance covariance and Pearson keep more of theirs. EXP6 shows that this weakness comes from the bandwidth. Each series has its own seasonal phase, so within a sample $|X|$ and $|Y|$ move together even under the null. The median-bandwidth kernel responds to this co-movement, which widens its null distribution. The median bandwidth itself is not inflated (0.94 against 0.95 for the Gaussian row). Doubling the bandwidth raises size-adjusted HSIC power from 16% to 43%, and quadrupling it gives 73%, close to Pearson. The same wide bandwidth cuts power against the squared alternative from 37% to 7%, because a wide Gaussian kernel approaches a squared Pearson correlation. No fixed multiple of the median is uniformly good. Circular shifts held size at every bandwidth we tried, between 4.0% and 5.8%.

4.5 Seasonal adjustment and slow scale drift

Table 6 reruns the seasonal nulls after adjustment. Removing period-bin means fixes the seasonal-mean failure. At $n = 500$ and $H = 0.5$, HSIC with wild multipliers falls from 81.5% to 8.6%. It does not fix the seasonal-scale failure. Removing only the mean leaves HSIC at 73.8% at $n = 2000$. Dividing also by period-bin standard deviations brings it to 7.2%, close to the adjusted Gaussian row (6.3%). Estimating the period works as well. The periodogram recovers the true period in every seasonal series we checked. At $H = 0.8$, the adjusted seasonal-scale rows return to the level of the adjusted Gaussian rows. What remains is the memory effect.

Table 6: EXP4 null rejection percentages at $n = 2000$ after seasonal adjustment. A0 is unadjusted (EXP2). A1 removes period-bin means. A2 also divides by period-bin standard deviations. SEAS-V-slow has period $4n$ and cannot be bin-adjusted. Every cell has 1,000 replications.

		HSIC wild		HSIC shift		HSIC sub		dCov sub		Pearson NW	
Generator	Adj.	.5	.8	.5	.8	.5	.8	.5	.8	.5	.8
G0 Gaussian	A0	6.8	8.1	5.3	5.3	5.8	17.5	6.8	22.8	5.0	14.1
G0 Gaussian	A2	6.3	11.5	5.1	6.8	5.6	23.9	5.9	26.2	5.8	15.8
SEAS-V seasonal scale	A0	73.3	68.0	5.0	5.1	71.5	73.0	45.6	53.3	4.7	13.4
SEAS-V seasonal scale	A1	73.8	63.2	4.3	5.2	71.7	69.3	46.8	46.3	6.0	14.8
SEAS-V seasonal scale	A2	7.2	9.7	5.0	4.7	6.3	21.9	5.8	26.4	5.8	15.3
SEAS-V-slow drift	A0	16.8	21.4	10.1	10.2	13.4	39.1	6.4	35.2	5.1	16.2

Adjustment has a cost under memory. On the Gaussian row at $H = 0.8$, removing 50 bin means and scales raises HSIC subsampling from 17.5% to 23.9%. The slow drift is the harder case. A seasonal scale with period $4n$ shows only a quarter cycle in each window, so bin adjustment cannot remove it. With no memory, it raises HSIC with wild multipliers to 16.8% and circular shifts to 10.1% at $n = 2000$. At $n = 500$ the same rows are 8.8% and 6.2%, so the excess grows with the window length. A shift cannot re-randomize a pattern that the window sees only once. With 1,000 replications, the standard error of these rates is about 1 point.

4.6 Stock durations: an applied boundary test

The six-month extraction gives 124 trading days per stock and 9.68 million AAPL, 6.57 million MSFT, and 14.21 million NVDA positive durations. Every stock has observations in all 26 clock

bins. The paired windows cannot overlap in calendar time because they come from separated days. Separation alone does not make the underlying processes independent.

At $n = 2000$, HSIC circular shifts reject 3.4%–5.6% of the sampled pairs, wild multipliers reject 4.8%–8.6%, and subsampling rejects 7.2%–10.2% (Table 7). The table gives each stock pairing, including the AAPL–AAPL check. At $n = 500$, wild-multiplier rates are 8.6%–11.0% and subsampling rates are 9.4%–11.6%. There is no consistent rise with window size across these cells.

The smooth diurnal adjustments leave these fractions in place. Under D2, no stock cell moves by more than 0.6 points at any window size, and the original adjustment reproduces every saved row exactly. D2 does remove the time-of-day scale pattern: the ratio of the largest to the smallest diurnal standard deviation falls from about 1.4 to 1.01. What D2 does not remove is the variance trend inside each window. We split every $n = 2000$ window into eight segments and regressed log segment variance on position. The slope exceeds the 95th percentile of its own permutation distribution in 45.6% of windows before adjustment and 43.7% after, against 5% with no trend. Pairs with a trend in both windows reject more often. HSIC subsampling rejects 11.0% of them, against 7.5% of pairs with no trend in either window. The drift is therefore local to the window, not a time-of-day effect, and it matches the slow-drift limit of Section 4.5.

Table 7: Stock duration placebo rejection percentages at $n = 2000$ and a nominal 5% level. Each row has 500 pairs. The processes' independence is unverified and days are reused, so these are descriptive fractions rather than false-alarm estimates.

Pairing	HSIC wild	HSIC shift	HSIC sub	dCov wild	dCov sub
AAPL–MSFT	8.6	5.6	8.4	8.4	8.8
AAPL–NVDA	5.8	4.8	9.2	6.0	7.2
MSFT–NVDA	7.2	5.4	10.2	6.4	11.2
AAPL–AAPL	4.8	3.4	7.2	5.0	7.6

The trade records also limit what we can test. On a typical day, about one fifth of AAPL trades, one quarter of MSFT trades, and more than one third of NVDA trades have unspecified aggressor side, so we do not analyze trade signs. Days recur across sampled pairs; one day is used as many as 9–12 times on one side of a cell. Market regimes may persist across the 20-session gap. We cannot interpret Table 7 as a causal effect or a measured 5% false-alarm check.

5 Discussion

The simulations establish rule-specific false-alarm rates under their stated null generators. They show excess rejection at a memory sum of 1.4, below the covariance balance at 1.5. That balance predicts a change in the summability of a leading product feature; it does not determine a usable critical value. Circular shifts are closest to 5% on this grid, while subsampling is strongly oversized in several cells. We cannot extend either observation to other long-memory processes or changing market conditions without further work.

Theory and experiment meet only within stated assumptions. The subsampling theorem discussed in Section 2 may apply to restricted independent-fGn distance-covariance settings once our statistic and block rule are checked exactly. We still lack the corresponding HSIC null limit and a proof for its window rule. The MMD feature-rank argument likewise does not validate its multiplier or self-normalized critical values.

The stock study shows how the rules behave on observed market durations, where a true process-independence null is unavailable. The 20-session gap prevents overlap but cannot turn a rejection into a measured false alarm. A stronger claim would require a specified null, a validated rule, and matched surrogates. EXP2 separates memory from heavy tails, volatility, nonlinear

dynamics, and seasonality in simulation, and its seasonal-scale result bears on the stock arm. The main stock arm removes a clock-bin mean from log durations but not a clock-bin scale. A window of 2,000 durations covers about 7 to 15 minutes of a 390-minute session, so it sees only a small part of the daily cycle. In simulation, a slow scale drift of this kind raises HSIC with wild multipliers to 16.8% at $n = 2000$ with no memory (Section 4.5). A smooth diurnal scale adjustment removes the daily pattern but leaves the within-window trend, and pairs with that trend reject more often (Section 4.6). This links the stock fractions to slow drift, but it does not prove that drift causes them.

EXP2 separates two failures that a user would see as the same false alarm. The first comes from memory. Process features change its size, but not its direction, except stochastic volatility, which moves memory into a layer the independence tests barely weight. The second comes from seasonality. A shared seasonal mean breaks every single-path test, so its remedy is deseasonalization or a shift-based rule, not a different kernel. A shared seasonal scale breaks HSIC and distance covariance while leaving Pearson at 5%. That failure is specific to tests that look beyond linear correlation. On this grid, circular shifts were the only rule that held size under both seasonal forms. Section 4.4 shows that this is not bought with lost power. Section 4.5 shows that estimated adjustment is a second remedy, provided it removes the seasonal scale and not only the mean.

Kernel choice is a second lever, alongside the critical-value rule. The E2 results suggest that features without a rank-one component hold size until $H = 3/4$. A kernel or feature map built to remove the first Hermite layer might therefore postpone over-rejection. The same layer, however, carries the first-order effect of a mean shift. For a feature g of a standard Gaussian ξ, $\partial_\mu \mathbb{E}g(\xi + \mu)|_{\mu=0} = \mathbb{E}\{\xi g(\xi)\}$, which is the first Hermite coefficient of g. A feature with no rank-one component therefore responds to a small mean shift only at second order. Whether a kernel can trade some of this sensitivity for size control, and at what cost in power, is open. Answering it requires size and size-adjusted power on the same cells, which this paper reports for the existing rules (Section 4.4) but not for a kernel built to remove the first layer.

Three proof problems remain. First, HSIC needs a null limit that covers transformed features and mixed-memory pairs. Second, an H-free MMD rule needs a specified statistic and block construction, a size proof, and a power comparison. Third, any claim that local asymmetry changes Hermite rank needs an explicit transform, kernel, and asymmetry sequence. These are open questions, not results of this paper.

6 Limitations

Our main simulation map covers Gaussian and Gaussian-subordinated series and one discrete slow-mean-reversion comparison. EXP2 adds four processes outside the Hermite framework, but at one parameter setting each, with 1,000 replications per cell and $n \leq 2000$. Its seasonal generators use one period, one amplitude, and a uniform random phase. Most kernel cells stop at $n = 2000$. The $n = 4000$ cells cover only equal-memory pairs, and $n = 8000$ has 144 replications per cell. Adjacent-grid slopes are noisy finite differences, not estimates of a structural threshold. We vary the bandwidth only by fixed multiples of the median, and we did not study data-driven bandwidth choice. The power study uses four alternatives, at one or two effect sizes each. Some rules already miss 5% at the $H = 0.5$ baseline.

The market illustration covers three stocks and six months. Reused days and changing stock-specific regimes make its pair fractions dependent. Pooled smooth adjustment of the mean and scale leaves the stock fractions unchanged, but it cannot remove scale drift inside a window, and we have no per-window remedy for it. Raw gaps and timestamps remain in the local cache for further checks. No theorem in Section 2 establishes validity for the exact implemented HSIC or MMD rules or for the stock-window comparison.

7 Conclusion

For independence tests on persistent series, we recommend one rule. Remove the seasonal mean and the seasonal scale from each series. Then test with HSIC and circular shifts. In our simulations at $n = 2000$, this rule rejected 3.8%–7.0% of true nulls across the memory grid and the process features. At matched size it lost at most 4.9 points of power to the best other HSIC rule in 18 cells. Its good size therefore does not come from rarely rejecting.

The rule has three limits. First, no long-memory theory covers circular shifts. Under strong memory their rate rises slowly with n, from 5.7% at $n = 2000$ to 6.3% at $n = 4000$ at a memory sum of 1.6. Second, a slow scale drift that a window sees only once defeats it. Such a drift raised circular shifts to 10.1% at $n = 2000$, and no per-window adjustment removed it. Third, seasonal adjustment costs size under memory. At $H = 0.8$ it raised HSIC subsampling from 17.5% to 23.9%. The rule's size did not depend on the kernel bandwidth, but its power did.

Two findings lie behind the rule. The first is that the critical-value rule and the feature rank set the false-alarm rate. Rank-one split MMD over-rejects at weaker memory than the even rank-two transforms. For independent pairs, HSIC and distance-covariance rejection depends strongly on the rule, even below a memory sum of 1.5, and the excess does not shrink as n grows to 4000. Hermite rank explains why these patterns differ, but it does not supply a valid cutoff. The second finding is that a seasonal scale shared by two independent series misleads kernel tests without any memory. Pearson correlation is unaffected. A shared seasonal mean with random phase misleads every single-path test, and Proposition 1 proves that its size tends to one for the common rules. Removing only the seasonal mean leaves the seasonal-scale failure in place.

Split MMD has no comparable rule. No tested critical-value rule held its size under memory. Memory also removes power against a fixed mean shift, because the two half-means spread apart under the null. Proposition 2 shows that this loss is shared by every location-invariant test. At a shift scaled to the null spread, MMD power stays level or rises with memory. The stock-duration fractions show how the rules behave on market data. Without a known independence null, they remain descriptive.

Data and code availability. The public repository and reproduction paths are given in Appendix B. It contains simulation code and outputs, stock request manifests and derived results, and the manuscript source. Raw Databento trades are excluded. No market-data redistribution right is claimed.

Ethics, funding, and competing interests. This draft reports simulation and market-trade analyses, with no human-subject intervention. Funding, author contributions, AI-tool use, and competing-interest statements require author verification before submission.

A Proofs of Propositions 1 and 2

A.1 Ergodic averages with periodic weights

Write $W_t = (\xi_t, \zeta_t)$. Let $\gamma(k) = \frac{1}{2}(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H})$ be the fGn autocovariance. For $H \neq 1/2$, $\gamma(k) \sim H(2H-1)k^{2H-2}$, and for $H = 1/2$, $\gamma(k) = 0$ when $k \neq 0$. In both cases $\gamma(k) \to 0$ because $H < 1$.

Lemma 1. *Fix $L \geq 0$ and $P \geq 1$. Let $f_0, \ldots, f_{P-1}$ be measurable functions of $(W_t, \ldots, W_{t-L})$ with $\mathbb{E}|f_j(W_L, \ldots, W_0)| < \infty$. Then, almost surely,*

$$\frac{1}{n} \sum_{t=L}^{n+L-1} f_{t \bmod P}(W_t, \ldots, W_{t-L}) \longrightarrow \frac{1}{P} \sum_{j=0}^{P-1} \mathbb{E} f_j(W_L, \ldots, W_0).$$

Proof. A stationary Gaussian sequence whose covariance tends to zero is mixing under the shift [Maruyama, 1949]. See also Cornfeld et al. [1982, Chapter 14]. So the shifts of ξ and of ζ are mixing. The two sequences are independent, and the direct product of two mixing systems is mixing [Walters, 1982]. Hence the shift S of W is mixing. Then S^P is mixing, because $\Pr(S^{-Pk}A \cap B) \to \Pr(A)\Pr(B)$ is a subsequence of the mixing limit for S. A mixing transformation is ergodic. For each residue j, Birkhoff's theorem applied to S^P and to f_j evaluated at the window ending at time $j+L$ gives $K^{-1}\sum_{k<K} f_j(W_{j+L+kP}, \dots, W_{j+kP}) \to \mathbb{E}f_j$ almost surely. The sum in the lemma splits into these P residue classes. Each class holds $n/P + O(1)$ terms. Averaging the P limits with weights tending to $1/P$ gives the claim. □

Two special cases are used repeatedly. Taking $f_j(w) = g_j\, f(w)$ gives $n^{-1}\sum_t g_t f(W_t) \to \bar{g}\,\mathbb{E}f(W_0)$ for a P-periodic sequence g with mean $\bar{g}$. Taking $f_j(w) = h\{w + (s_j, q_j)\}$ gives limits of empirical averages of a periodically shifted sequence.

A.2 Proof of Proposition 1(i)

The phases (U, V) are independent of W and take finitely many values. It is therefore enough to prove almost-sure convergence for each fixed pair (u, v). Write $s_t = a\sin\{2\pi(t+u)/P\}$ and $q_t = a\sin\{2\pi(t+v)/P\}$. For $P \geq 3$ the trigonometric sums give

$$\frac{1}{P}\sum_{j=0}^{P-1} s_j = 0, \qquad \frac{1}{P}\sum_{j=0}^{P-1} s_j^2 = \frac{a^2}{2}, \qquad \frac{1}{P}\sum_{j=0}^{P-1} s_j q_j = \frac{a^2}{2}\cos\frac{2\pi(u-v)}{P}.$$

The last identity uses $\sin\alpha\sin\beta = \frac{1}{2}\{\cos(\alpha-\beta) - \cos(\alpha+\beta)\}$. The sum of $\cos(\alpha_j + \beta_j)$ vanishes because $e^{4\pi i/P} \neq 1$ for $P \geq 3$. Lemma 1 with $L = 0$ gives, almost surely, $\bar{X} \to 0$ and $\bar{Y} \to 0$. It also gives

$$\frac{1}{n}\sum_t X_t^2 = \frac{1}{n}\sum_t \xi_t^2 + \frac{2}{n}\sum_t s_t\xi_t + \frac{1}{n}\sum_t s_t^2 \to 1 + 0 + \frac{a^2}{2}.$$

The same holds for Y. For the cross moment, $n^{-1}\sum_t \xi_t\zeta_t \to \mathbb{E}\xi_0\zeta_0 = 0$, and the terms $n^{-1}\sum_t s_t\zeta_t$ and $n^{-1}\sum_t q_t\xi_t$ tend to 0. Hence $\hat{C}_n \to (a^2/2)\cos\{2\pi(u-v)/P\}$ and $r_n \to c(u-v)$.

The gap $D = (U - V) \bmod P$ is uniform because U is uniform and independent of V. So $\mathbb{E}\cos(2\pi D/P) = P^{-1}\sum_d \cos(2\pi d/P) = 0$. Finally, for $0 \leq d < P$, $\cos(2\pi d/P) = 0$ exactly when $d = P/4$ or $d = 3P/4$. Such integers exist only when 4 divides P. □

A.3 Proof of Proposition 1(ii)

Let $C(D) = (a^2/2)\cos(2\pi D/P)$, so $\hat{C}_n \to C(D)$ almost surely by part (i). The variable D takes finitely many values. Choose $\eta > 0$ below $\frac{1}{2}\min\{|C(d)| : C(d) \neq 0\}$. On $\{C(D) \neq 0\}$, the test fails to reject only if $|\hat{C}_n| \leq 2\eta$ or $z_n(\hat{v}_n/n)^{1/2} \geq 2\eta$. The first event has probability tending to zero by almost-sure convergence. The second has probability tending to zero because z_n is bounded and $\hat{v}_n/n \to 0$ in probability. Hence $\Pr(\text{no rejection}, C(D) \neq 0) \to 0$. Since $c(D) \neq 0$ exactly when $C(D) \neq 0$, this gives the bound.

It remains to place the two long-run variance rules in the class. Let $u_t = (X_t - \bar{X})(Y_t - \bar{Y})$, let $\tilde{u}_t = u_t - \bar{u}$, and let $\hat{\gamma}(j) = n^{-1}\sum_t \tilde{u}_{t+j}\tilde{u}_t$. By Cauchy–Schwarz, $|\hat{\gamma}(j)| \leq \hat{\gamma}(0)$. Expanding u_t^2 as a polynomial in $\bar{X}$, $\bar{Y}$, ξ_t, ζ_t, s_t, q_t and applying Lemma 1 shows that $\hat{\gamma}(0)$ converges almost surely to a finite limit.

Newey–West. The Bartlett estimate is $\hat{v}_n = \hat{\gamma}(0) + 2\sum_{1\leq j<S_n}(1 - j/S_n)\hat{\gamma}(j)$. It is nonnegative, because the Bartlett weights form a positive-definite sequence. Its weights lie in $[0, 1]$, so $\hat{v}_n \leq (2S_n + 1)\hat{\gamma}(0)$. Thus $\hat{v}_n/n \to 0$ whenever $S_n = o_p(n)$. The implemented bandwidth is the Andrews [1991] AR(1) plug-in $S_n = 1.1447(\hat{\alpha}_n n)^{1/3}$, with $\hat{\alpha}_n = 4\hat{\rho}_n^2/\{(1-\hat{\rho}_n)^2(1+\hat{\rho}_n)^2\}$ and $\hat{\rho}_n = \sum_t \tilde{u}_t\tilde{u}_{t-1}/\sum_t \tilde{u}_{t-1}^2$. By Lemma 1 with $L = 1$, $\hat{\rho}_n \to \rho^*$ almost surely. Suppose $|\rho^*| = 1$. Then $n^{-1}\sum_t(\tilde{u}_t \mp \tilde{u}_{t-1})^2 \to 0$. By Lemma 1 this limit equals $P^{-1}\sum_j \mathbb{E}F_j^2$, where

$F_j = (\xi_1 + s_j)(\zeta_1 + q_j) \mp (\xi_0 + s_{j-1})(\zeta_0 + q_{j-1}) + \kappa$ for a constant κ that comes from centring. The vector $(\xi_0, \xi_1, \zeta_0, \zeta_1)$ is nondegenerate Gaussian, because $|\gamma(1)| = |2^{2H-1} - 1| < 1$. F_j is a nonconstant polynomial in it, since it contains $\xi_1\zeta_1$ with coefficient one. So $\mathbb{E}F_j^2 > 0$, a contradiction. Hence $|\rho^*| < 1$, $\hat{\alpha}_n$ converges to a finite limit, and $S_n = O(n^{1/3})$ almost surely.

Equal-weighted cosine. The estimate is $\hat{v}_n = \nu_n^{-1} \sum_{j=1}^{\nu_n} \Lambda_j^2$ with $\Lambda_j = (2/n)^{1/2} \sum_t \cos\{\pi j(t - 1/2)/n\}\tilde{u}_t$. These cosine vectors, together with $n^{-1/2}\mathbf{1}$, form the orthonormal DCT-II basis of $\mathbb{R}^n$. So $\sum_{j \le \nu_n} \Lambda_j^2 \le \|\tilde{u}\|^2 = n\hat{\gamma}(0)$ and $\hat{v}_n/n \le \hat{\gamma}(0)/\nu_n \to 0$ when $\nu_n \to \infty$. The implementation uses $\nu_n = \lfloor 0.4n^{2/3} \rfloor$. Its Student-$t$ critical value is bounded by the $\nu = 1$ value. □

A.4 Proof of Proposition 1(iii)

Fix (u, v) as before. Let $\mu_n = n^{-1} \sum_t \delta_{(X_t, Y_t)}$ be the empirical measure, and define the Gaussian mixture

$$\mu_{u,v} = \frac{1}{P} \sum_{j=0}^{P-1} N((s_j, q_j)^\top, I_2).$$

Step 1: $\mu_n \to \mu_{u,v}$ weakly, almost surely. For fixed $\tau \in \mathbb{R}^2$, Lemma 1 with $f_j(w) = \exp\{i\tau^\top(w + (s_j, q_j))\}$ gives convergence of the characteristic function of μ_n at τ. Intersecting over rational τ gives one almost-sure event. On it, $n^{-1} \sum_t (X_t^2 + Y_t^2)$ also converges, so (μ_n) is tight. Every subsequential weak limit then has the characteristic function of $\mu_{u,v}$ on a dense set, hence everywhere by continuity. So every subsequential limit equals $\mu_{u,v}$.

Step 2: the bandwidths converge. The X-marginal of $\mu_{u,v}$ is $\mu^X = P^{-1} \sum_j N(s_j, 1)$. As a multiset, $\{s_j\}$ does not depend on u, and it equals $\{q_j\}$. Let G be the law of $|X - X'|$ for X, X' independent with law μ^X. It has a continuous, strictly increasing distribution function on $[0, \infty)$, so its median $\sigma^* > 0$ is unique. The empirical law of the off-diagonal distances $|X_s - X_t|$, $s \ne t$, is the image of $\mu_n^X \otimes \mu_n^X$ under $(x, x') \mapsto |x - x'|$, minus a diagonal mass $1/n$, renormalized. By Step 1 and the continuous mapping theorem, it converges weakly to G. Because G is continuous, the convergence is uniform in the argument, so the median-distance bandwidth $\hat{\sigma}_n \to \sigma^*$. The positive-distance fallback is eventually inactive. The same holds for $\hat{\tau}_n$, with the same limit σ^*.

Step 3: $V_n \to h(u - v)$. For bandwidths (σ, τ) and a probability ν on $\mathbb{R}^2$, write $k_\sigma(x, x') = \exp\{-(x - x')^2/(2\sigma^2)\}$ and

$$\mathrm{HSIC}_{\sigma,\tau}(\nu) = \mathbb{E}\, k_\sigma(X_1, X_2) l_\tau(Y_1, Y_2) - 2\,\mathbb{E}\, k_\sigma(X_1, X_2) l_\tau(Y_1, Y_3) + \mathbb{E}\, k_\sigma(X_1, X_2)\, \mathbb{E}\, l_\tau(Y_3, Y_4),$$

with (X_i, Y_i) independent draws from ν and $l_\tau = k_\tau$. The statistic of Section 3 satisfies $V_n = n^{-2} \operatorname{tr}(K^c L^c) = \mathrm{HSIC}_{\hat{\sigma}_n, \hat{\tau}_n}(\mu_n)$, the standard identity for the biased estimator [Gretton et al., 2005]. Each term is an integral of a bounded continuous function against $\nu^{\otimes r}$ with $r \le 4$. Weak convergence of μ_n implies weak convergence of $\mu_n^{\otimes r}$ [Billingsley, 1999, Theorem 2.8]. So $\mathrm{HSIC}_{\sigma^*,\sigma^*}(\mu_n) \to \mathrm{HSIC}_{\sigma^*,\sigma^*}(\mu_{u,v})$. Next, $|\partial_\sigma k_\sigma| \le (2/e)/\sigma$ uniformly in the arguments. Kernels are bounded by one, so $|\,\mathrm{HSIC}_{\sigma,\tau}(\nu) - \mathrm{HSIC}_{\sigma',\tau'}(\nu)| \le 4(\|k_\sigma - k_{\sigma'}\|_\infty + \|k_\tau - k_{\tau'}\|_\infty)$ for every ν. With Step 2 this gives $V_n \to h(u - v) := \mathrm{HSIC}_{\sigma^*,\sigma^*}(\mu_{u,v})$. The value depends on (u, v) only through $u - v$, by a shift of the index j.

Step 4: $h(d) > 0$ for every d when $P \ge 5$. The product $k_\sigma \otimes k_\tau$ is a Gaussian kernel on $\mathbb{R}^2$. It is characteristic [Sriperumbudur et al., 2010]. $\mathrm{HSIC}_{\sigma,\tau}(\nu)$ is the squared MMD between ν and $\nu^X \otimes \nu^Y$ for this kernel [Gretton et al., 2005]. So $h(d) = 0$ only if $\mu_{u,v}$ equals the product of its marginals. Let π be the law of (s_J, q_J) with J uniform on $\{0, \dots, P-1\}$. Then $\mu_{u,v} = \pi * N(0, I_2)$ and $\mu^X \otimes \mu^Y = (\pi^X \otimes \pi^Y) * N(0, I_2)$. The Gaussian characteristic function has no zeros, so the two are equal only if $\pi = \pi^X \otimes \pi^Y$. Suppose this holds. Let $x^* = \max_j s_j$. The equation $a \sin\theta = x^*$ has at most two solutions $\theta \in [0, 2\pi)$, so at most two indices j have $s_j = x^*$. Under π, the point x^* is paired with at most two values of q. Under a product measure it is paired with every point of the support of π^Y. So that support has at most two points. But q_j takes at

least $\lceil P/2 \rceil \geq 3$ distinct values, since each value is taken at most twice. This is a contradiction. Hence $h(d) > 0$ for every d, and $\min_d h(d) > 0$ since d ranges over a finite set.

Step 5: the rules. On every path, $T_n/n = V_n \to h(D) > 0$ almost surely. If $\kappa_n/n \to 0$ in probability, then $\Pr(T_n \leq \kappa_n) \leq \Pr\{V_n \leq h(D)/2\} + \Pr\{\kappa_n/n \geq h(D)/2\} \to 0$, using $h(D) \geq \min_d h(d) > 0$.

Double-centred Gaussian Gram entries satisfy $|K^c_{ij}| \leq 2$. So $M = K^c \circ L^c$ has entries bounded by 4, and it is positive semidefinite by the Schur product theorem.

Subsampling. Each window statistic is $T_\ell = \ell^{-1} \sum_{i,j \leq \ell} K^c_{ij} L^c_{ij} \leq 4\ell$. This holds for any bandwidth, including the full-sample bandwidth that the rule reuses. Its 95% quantile is at most $4\ell_n = o(n)$.

Wild multipliers. Let w be block-Markov Rademacher with $\mathrm{Cov}(w_s, w_t) = R_{st} = e^{-|s-t|/\ell_n}$, and $\tilde{w} = Cw$ with $C = I - n^{-1}\mathbf{1}\mathbf{1}^\top$, or $\tilde{w} = w$ for the raw rule. Given the data, $T^* = n^{-1}\tilde{w}^\top M \tilde{w} \geq 0$, and

$$\mathbb{E}_w T^* = n^{-1} \operatorname{tr}(CMC\,R) \leq n^{-1} \lambda_{\max}(R) \operatorname{tr}(CMC) \leq n^{-1}(1 + 2\ell_n) \operatorname{tr}(M) \leq 4(1 + 2\ell_n).$$

Here $\lambda_{\max}(R) \leq \max_s \sum_t R_{st} \leq 1 + 2\ell_n$, $\operatorname{tr}(CMC) \leq \operatorname{tr}(M)$ because $\mathbf{1}^\top M \mathbf{1} \geq 0$, and $M_{ii} \leq 4$. With B draws, the rule rejects when $(1 + \#\{b : T^*_b \geq T_n\})/(B+1) \leq \alpha$, that is, when at most $k_B = \lfloor \alpha(B+1) \rfloor - 1$ draws satisfy $T^*_b \geq T_n$. We assume $\alpha(B+1) \geq 1$, so $k_B \geq 0$. For $B = 500$ and $\alpha = 0.05$, $k_B = 24$. By Markov's inequality each draw exceeds T_n with conditional probability $p_n \leq 4(1 + 2\ell_n)/T_n$. So the conditional probability of no rejection is at most Bp_n. Since $T_n/\ell_n \to \infty$ almost surely when $\ell_n = o(n)$, dominated convergence gives $\Pr(\text{reject}) \to 1$. The implementation uses $\ell_n = \lfloor n^{1/2} \rfloor$ for both rules. □

Remark 1 (Circular shifts, proof sketch). The shift rule compares T_n with $T_n(s) = n\,\mathrm{HSIC}(\mu_n^{(s)})$, where $\mu_n^{(s)}$ is the empirical measure of $(X_t, Y_{t+s \bmod n})$, for s in a range S_n of about $0.8n$ offsets. When P divides n, as for $n = 2000$ and $P = 50$, the shifted seasonal component of Y is exactly a sinusoid with phase $V + s$. The shifted pair is then a SEAS-M pair with gap $D - s$, and its statistic is also of order n. Suppose that $T_n(s)/n \to h(D - s)$ uniformly over $s \in S_n$. Then the leading term of the p-value is the fraction of offsets with $h(D - s) \geq h(D)$. That fraction tends to $\pi(D) = P^{-1}\#\{d : h(d) \geq h(D)\}$, because S_n covers each residue class modulo P about equally often. Since D is uniform, $\Pr\{\pi(D) \leq \alpha\} \leq \alpha$. So the phase gap does not drive the size to one, as it does in part (iii). Two steps are missing. First, uniform convergence over S_n is not proved. Each offset also re-pairs the noise, and the shifted noise has one wrap-around break. Second, h has ties. For even P, $h(d) = h(P/2 - d)$, because the two limit measures differ by the reflection $y \mapsto -y$. Among offsets in a tied class, the decision is settled by lower-order terms, where the long-memory noise matters. The sketch therefore explains why the size is near α rather than near one. It does not prove that the size equals α.

Remark 2 (Distance covariance). The same steps apply to distance covariance with subsampling, with two changes. The distance kernel is unbounded, so Step 3 needs uniform integrability of $|X_s - X_t|$ under $\mu_n^{\otimes 2}$. This follows from almost-sure convergence of $n^{-1}\sum_t X_t^2$. The window bound $T_\ell \leq 4\ell$ becomes a bound by ℓ times the largest window second moment. That moment grows at most logarithmically for Gaussian noise. We have not written out these two steps, so this case is stated without proof.

A.5 Proof of Proposition 2

(i) Let $S_k = \sum_{t \leq k} \xi_t$. Unit-variance fGn has $\mathrm{Var}(S_k) = k^{2H}$ exactly. Write $A = S_m$ and $B = S_{2m} - S_m$, so $\mathrm{Var}(A) = \mathrm{Var}(B) = m^{2H}$ by stationarity. From $\mathrm{Var}(A + B) = (2m)^{2H}$ we get $2\,\mathrm{Cov}(A, B) = (2m)^{2H} - 2m^{2H}$. Hence

$$\mathrm{Var}(A - B) = 2m^{2H} - 2\,\mathrm{Cov}(A, B) = 4m^{2H} - (2m)^{2H},$$

and $\Delta_m = (A - B)/m$ gives (7). The ratio R_m is $\delta/\operatorname{Var}(\Delta_m)^{1/2}$.

(ii) The first-half mean minus the second-half mean of Z equals $\Delta_m - \delta$. It is Gaussian with mean $-\delta$ and variance $\operatorname{Var}(\Delta_m)$, and μ cancels. The power formula is the two-sided normal tail at noncentrality R_m.

(iii) Let L be any $(n-1) \times n$ matrix of rank $n-1$ with $L\mathbf{1} = 0$. The statistic LZ is a maximal invariant under $Z \mapsto Z + b\mathbf{1}$. So every invariant test is a function of LZ [Lehmann and Romano, 2005, Theorem 6.2.1]. Its law is $N(\delta Lc, \Sigma)$ with $\Sigma = L\Gamma L^\top$, which does not involve μ. By the Neyman–Pearson lemma, the most powerful level-α test of $\delta = 0$ against a fixed $\delta > 0$ in this family rejects for large $(Lc)^\top \Sigma^{-1} LZ$. Its power is $1 - \Phi\{z_{1-\alpha} - \delta(c^\top L^\top \Sigma^{-1} Lc)^{1/2}\}$. Any invariant level-$\alpha$ test has at most this power.

It remains to show $L^\top (L\Gamma L^\top)^{-1} L = \Gamma^{-1} - \Gamma^{-1}\mathbf{1}(\mathbf{1}^\top\Gamma^{-1}\mathbf{1})^{-1}\mathbf{1}^\top\Gamma^{-1}$. Multiply the left side by $\Gamma^{1/2}$ on both sides. The result is the orthogonal projection onto the range of $\Gamma^{1/2}L^\top$. That range has dimension $n-1$ and is orthogonal to $\Gamma^{-1/2}\mathbf{1}$, since $L\Gamma^{1/2}\Gamma^{-1/2}\mathbf{1} = L\mathbf{1} = 0$. So the projection equals $I - \Gamma^{-1/2}\mathbf{1}\mathbf{1}^\top\Gamma^{-1/2}/(\mathbf{1}^\top\Gamma^{-1}\mathbf{1})$. Multiplying back by $\Gamma^{-1/2}$ on both sides gives the identity, and hence $c^\top L^\top \Sigma^{-1} Lc = I_n$. By the Gauss–Markov theorem for the regression of Z on $(\mathbf{1}, c)$ with covariance Γ, I_n^{-1} is the variance of the generalized-least-squares estimator of δ. □

Since that estimator is the best linear unbiased one, $I_n \geq 1/\operatorname{Var}(\Delta_m)$, and the cap in (iii) is never below the power in (ii). Part (iii) assumes that H and the unit variance are known. A test that controls size over a range of H is also level α at each H in the range, so the cap applies to it too.

B Reproducibility and audit trail

The public GitHub repository contains the code and saved outputs named below. All paths are relative to its root. Raw Databento files are excluded and require separate licensed access. The stock manifests give source-file hashes, dates, and API cost quotes. The stock experiment README gives the commands to download, extract, test, and summarize those data.

Simulation outputs. The initial experiments are in two `experiments/` folders ending `mmd-dml` and `kernel-linear`. The `EXP1-20261016` folder holds the H5, E2, and E3 scripts, cell files, and summaries. Its manifests record seeds and target counts. The `EXP2-20261023` folder holds the process-feature design record, generator checks, pilot, main-run cell files, `summary_main.csv`, and `RESULTS.md`. The design record was written before the pilot and lists two later amendments. The `EXP3-power-20261025` and `EXP4-deseason-20261025` folders hold the power and seasonal-adjustment studies in the same format, each with its design record, amendments, pilot, main-run cells, and tables. The `EXP5-scaled-n-20261025` and `EXP6-bandwidth-20261025` folders follow the same format.

The initial MMD cells used 2,000 null replications up to $n = 2000$ and 1,000 at $n = 4000$ after a runtime fallback. E2 used 2,000 null replications up to $n = 1000$ and 1,000 at $n = 2000$; each power cell used 500. E3 used 2,000 per cell. The paper's `make_figures.py` reads the saved E2/E3 summaries and records input and output hashes in `figures/manifest.json`. It does not rerun simulations.

Self-normalized block statistic. Let $S_{a:b} = \sum_{t=a}^{b} \phi(Z_t)$ and $m = \lfloor n/2 \rfloor$. The denominator in `mmdcal.py` measures how far feature partial sums deviate from a straight-line bridge within each half:

$$V_n = \frac{1}{n^2}\left\{\sum_{k=1}^{m}\left\|S_{1:k} - \frac{k}{m}S_{1:m}\right\|^2 + \sum_{j=1}^{n-m}\left\|S_{n-j+1:n} - \frac{j}{n-m}S_{m+1:n}\right\|^2\right\}. \tag{9}$$

The test uses $n\,\mathrm{MMD}_n^2 / V_n$. To find a critical value, it computes the same ratio in every contiguous block of length $\lfloor n^{0.5} \rfloor$ or $\lfloor n^{0.7} \rfloor$ and takes the empirical 95th percentile. Each block is split into halves. A change in block length changes the rule. Self-normalization alone does not prove that either choice has the intended size.

Multiplier control and decision record. The $H = 0.5$ control is in the repository's `investigations` directory; the reproduction guide gives its full path. At each sample size, centred and raw Markov multipliers were applied to the same 1,000 independent Gaussian pairs, with 500 multiplier draws per pair.

The selection rule required raw-multiplier rejection between 3% and 7% in all six test-by-size cells. It also required every raw-multiplier cell to be at least as close to 5% as its centred counterpart. Raw multipliers rejected below 5% in all six cells and failed the second condition in two. The stock arm therefore retained the centred implementation. The interpretation decisions dated 18, 19, and 22 October are in `FINDINGS.md`; they are not validation results.

Stock extraction and pair check. `experiments/STOCK-20261023/` contains request manifests, an extraction summary, code, 12 pair CSVs, and `stock_placebo_summary.json`. The summary script checks pair IDs, test status, day gaps, and within-day boundaries. All 6,000 saved pairs passed those checks; none had fewer than 20 trading sessions between windows. Raw trades and local duration caches are excluded from version control. The pair results remain descriptive because process independence is unknown and days recur across pairs.

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